

Andrey Veprikov

ARTIFICIAL INTELLIGENCE RESEARCHER · PHD STUDENT AT MIPT

☎ +7(912)759-62-52 | ✉ veprikov.as@phystech.edu | 📅 December 21st, 2001 | 🌐 github.com/Veprikov | 📧 @shkodnik1917 | 🎓 Andrey Veprikov

About Me

Researcher in artificial intelligence and optimization, PhD student at MIPT. Author of 15+ research works: **5 papers at A*/A venues** (AAAI, ACL, ICML, UAI) and **6 peer-reviewed journal articles**, including Q1. Research interests: efficient fine-tuning of large language models (LoRA, PEFT), next-generation optimizers (Muon, SignSGD), zero-order methods, and distributed and federated optimization. I combine research with applied industry work: **I lead a practical AI direction at Sber** (Russia's largest bank), rising from intern to head in under a year, leading a team and shipping ML solutions into the Bank's products. **Golden Reviewer at ICML**, reviewer for NeurIPS and the TMLR journal. Google Scholar: 50+ citations, h-index 5.

Education

Moscow Institute of Physics and Technology (MIPT)

Moscow, Russia

PhD Student, Computer Science

2026 – Present

- Phystech School of Applied Mathematics and Informatics (FPMI), Department of Mathematical Foundations of Control (MFC)
- Research focus: optimization methods and efficient training of large models

Moscow Institute of Physics and Technology (MIPT)

Moscow, Russia

M.Sc., Computer Science

Sep 2024 – Jun 2026

- Phystech School of Applied Mathematics and Informatics (FPMI), Department of Intelligent Systems (DIS)
- Diploma with Honors (Distinction), GPA **9.5/10**
- Thesis: optimal control of feedback loops in continual learning systems

Moscow Institute of Physics and Technology (MIPT)

Moscow, Russia

B.Sc., Applied Mathematics and Physics

Sep 2020 – Jun 2024

- Phystech School of Applied Mathematics and Informatics (FPMI), Department of Intelligent Systems (DIS)
- GPA **8.96/10**
- Thesis: a dynamic-systems model of the feedback loop effect in machine learning systems

Experience

MBZUAI, ML Department

Abu Dhabi, UAE

Research Assistant

Feb 2025 – Apr 2025

- International research experience at one of the world's leading AI universities
- Collaborated with a top international team (M. Takáč, S. Horváth) on efficient training of matrix parameters (Muon-like) and invariant optimization
- The collaboration produced a series of joint papers on optimizers (OPLoRA, Preconditioned Norms, LionMuon)

BRAIn Lab, MIPT

Moscow, Russia

Research Engineer (started as Technician)

Sep 2023 – Present

- Formally: Labs of Mathematical Optimization Methods, MIPT-Yandex, and Fundamental AI Research
- Pretraining of large language models and efficient optimization methods for it
- Next-generation optimizers (Muon, SignSGD), zero-order methods, warm-up scheduling
- AI Scientist: autonomous agents for conducting scientific research; research seminars

Sber (Russia's #1 bank) — Center for Practical AI / AI Lab

Moscow, Russia

Head of Direction (started as Intern)

Sep 2024 – Present

- Promoted from intern to head of direction at the Center for Practical AI
- Efficient fine-tuning of large language models; design of LoRA adapters (WeightLoRA, ACL 2026)
- Deployment of AI solutions into the Bank's product tasks

ISP RAS — Research Center for Trusted AI

Moscow, Russia

Senior Lab Researcher (started as Lab Assistant)

Feb 2024 – Present

- Development of federated learning and trusted AI methods
- Setting up and running distributed training of large language models

IITP RAS — Lab of Distributed Computing Systems

Moscow, Russia

Junior Research Scientist (started as Research Intern)

Feb 2024 – Oct 2025

- Studying feedback loops in machine learning systems
- Distributed and Markovian optimization methods

Skoltech

Moscow, Russia

Research Intern

Jun 2022 – Sep 2022

- Change-point detection for time series using machine learning

Key Publications

A*/A CONFERENCE PAPERS (AAAI, ACL, ICML, UAI)

Softsign: Smooth Sign in Your Optimizer For Better Parameter Heterogeneity Handling

Dmitrii Feoktistov, Timofey Belinsky, Andrey Veprikov, Amir Zainullin, Aleksandr Beznosikov

ICML 2026 (A*). *arXiv:2605.31371*, 2026

Leveraging Coordinate Momentum in SignSGD and Muon: Memory-Optimized Zero-Order Methods

Egor Petrov, Grigoriy Evseev, Aleksey Antonov, Andrey Veprikov, Pavel Plyusnin, Nikolay Bushkov, Stanislav Moiseev, Aleksandr Beznosikov

ICML 2026 Workshop. *arXiv:2506.04430*, 2026

Methods for Optimization Problems with Markovian Stochasticity and Non-Euclidean Geometry

Vladimir Solodkin, Andrey Veprikov, Alexander Chernyavskiy, Aleksandr Beznosikov

AAAI 2026 (A*). *arXiv:2408.01848*, 2026

Markovian Compression: Looking to the Past Helps Accelerate the Future

Andrey Veprikov, Vladimir Solodkin, Mikhail Rudakov, Petr Babkin, Aleksandr Beznosikov

UAI 2026 (A). *arXiv:2601.05398*, 2026

WeightLoRA: Keep Only Necessary Adapters

Andrey Veprikov, Vladimir Solodkin, Alexander Zyl, Andrey Savchenko, Aleksandr Beznosikov

ACL 2026, Main Track (A*). *arXiv:2506.02724*, 2026

PEER-REVIEWED JOURNAL ARTICLES Q1/Q2

A Mathematical Model of the Hidden Feedback Loop Effect in Machine Learning Systems

Andrey Veprikov, Alexander Afanasiev, Anton Khritankov

Knowledge and Information Systems (Q2) (2025). Springer, 2025

New aspects of black box conditional gradient: Variance reduction and one point feedback

Andrey Veprikov, Alexander Bogdanov, Vladislav Minashkin, Aleksandr Beznosikov

Chaos, Solitons & Fractals (Q1) 189 (2024) p. 115654. Elsevier, 2024

Zero Order Algorithm for Decentralized Optimization Problems

Andrey Veprikov, Egor Petrov, Grigoriy Evseev, Aleksandr Beznosikov

Doklady Mathematics (Q2) 110.1 (2024) S261–S277. Springer, 2024

Non-smooth setting of stochastic decentralized convex optimization problem over time-varying graphs

Aleksandr Lobanov, Andrey Veprikov, Georgiy Konin, Aleksandr Beznosikov, Alexander Gasnikov, Dmitry Kovalev

Computational Management Science (Q2) 20.1 (2023) p. 48. Springer, 2023

Projects & Open Source

Admin & contributor of the lab's GitHub organization

Git, CI/CD, GitHub Actions

BRAIn Lab

2024 – Present

- Administer the lab's GitHub organization; publish and maintain reproduction code for the lab's papers

Benchmarking framework for optimization methods

Python, PyTorch

LIB

2024 – Present

- Unified platform to compare optimizers (AdamW, SOAP, Shampoo, Muon, etc.) across CV, NLP, and LLM fine-tuning
- Flexible experiment configuration with PEFT and GLUE support; my most popular original repository

Open-source contribution

Python

llm-baselines

2024 – Present

- Develop a fork of EPFL's epfml/llm-baselines for LLM pretraining: add next-generation optimizers and experiment pipelines, benchmark methods on pretraining, contribute back to open source

AI assistant for the research cycle

Claude skills, Python

claude-brainlab

2025 – Present

- Custom build of 50+ skills and agents for Claude Code automating the entire research cycle
- arXiv→Zotero→Obsidian paper ingestion, reviewing, paper and slide preparation, knowledge-graph memory

Official code for the ACL 2026 paper

Python, HuggingFace

WeightLoRA

2025

- LoRA with automatic selection of critical adapters; experiments on DeBERTaV3, Llama 3, Qwen (GLUE, GSM8K, SQuAD)
- Submitted as a PR to the official PEFT library (under review) — fingers crossed for approval

Zero-order optimization library

Python, PyTorch

zero-order-optimization

2025

- A collection of gradient-free (zero-order) optimization methods in a torch-compatible format

ML libraries, Intelligent Systems Dept. (MIPT)

Python, PyTorch

Relaxit & HippoTrainer

2025

- Relaxit — relaxation methods for discrete distributions; the most-starred repository of the Intelligent Systems Dept. org (> 290 repositories), where I am an active contributor
- HippoTrainer — automatic hyperparameter optimization for PyTorch training

MIPT

LaTeX

M.Sc. & B.Sc. Theses

2024, 2026

- Optimal control of feedback loops in continual learning systems
- A dynamic-systems model of the feedback loop effect in AI systems

Applied ML pipeline

Python, vLLM

LLM Quantization & Inference Acceleration

2025

- Qwen3-8B quantization (SmoothQuant + GPTQ), vLLM deployment: —42% memory, +46% inference speed, quality preserved

Academic Service & Reviewing

2026 **Golden Reviewer**, ICML — one of the world's leading machine learning conferences

2026 **Reviewer**, NeurIPS — A* machine learning conference

2026 **Reviewer**, Transactions on Machine Learning Research (TMLR)

Awards & Scholarships

2023 – Present **Enhanced State Academic Scholarship**, For contribution to numerical optimization methods

MIPT

2023 – Present **K.V. Rudakov Named Scholarship**,

MIPT

2024 **Certificate of Honor**, For high achievements in science

MIPT

2022 – 2023 **Abramov Scholarship**, For academic excellence

MIPT

2020 **Gold Medal**, High school graduation with honors

Izhevsk

2020 **Grant for Gifted Students**, Udmurt Republic

Izhevsk

Skills

Programming Python, C/C++, SQL; algorithms and data structures

ML / DL PyTorch, HuggingFace, TensorFlow, NumPy, Pandas; LLM fine-tuning, LoRA/PEFT, distributed training

Mathematics Numerical optimization, probability theory, linear algebra, mathematical analysis

Tools Linux, Git, ~~TeX~~, Weights & Biases, vLLM, Slurm/tmux

Languages Russian (native), English (C1)

Additional Education

School on Tensor Methods in Mathematics and Data Sciences

Shenzhen, China

Shenzhen MSU-BIT University

November 2024

- Lectures on state-of-the-art results on tensor methods in modern artificial intelligence systems
- Presentation at the poster session

International School ASCOMP

Innopolis, Russia

Innopolis University

October, 2024

- Lectures on state-of-the-art results in numerical optimization methods
- Presentation at the poster session

School for Teaching Assistants

Moscow, Russia

Yandex

August 2024

- Lectures on modern concepts of learning
- Team work on projects for their further implementation in the educational programs of various universities

Control, Information and Optimization on Behalf of B.T. Polyak 2024

Novosibirsk, Russia

MIPT, HSE, MSU, Wildberries

July 2024

- School on optimization methods and machine learning
- Development of B.T. Polyak's ideas in modern science

Intensive on Reproducing State-of-the-art Scientific Results

Sochi, Sirius, Russia

Yandex

October 2023

- Lectures on modern scientific and engineering achievements in the IT sphere
- Teamwork on projects to reproduce experiments from scientific articles and infer their robustness

- School on optimization methods and machine learning
- Development of B.T. Polyak's ideas in modern science

Additional Publications

OTHER JOURNAL ARTICLES

Sampling of Semiorthogonal Matrices for the Muon Algorithm

Egor Petrov, Grigoriy Evseev, Aleksey Antonov, Andrey Veprikov, Nikolay Bushkov

Doklady Mathematics 112.3 (2025) pp. 572–580. Springer, 2025

On the properties of the limit set of a repeated machine learning process under feature space transformations

Andrey Veprikov, Anton Khritankov

Artificial Intelligence and Decision Making 1 (2025) pp. 56–66. 2025

CONFERENCE TALKS

An Out-of-Shelf Multi-Level Monte Carlo Approach for Average-Reward Reinforcement Learning

Alexander Chernyavskiy, Andrey Veprikov, Vladimir Solodkin, Aleksandr Beznosikov, Aleksandr Panov

International Conference on Computational Optimization (ICOMP), 2024

Adversarial Robustness through Wide Local Minima: A Simple Training Technique

Anton Khritankov, Andrey Veprikov, Sergey Smirnov

International Conference on Computational Optimization (ICOMP), 2024

Feature Space Transformations in the Repeated Machine Learning Process Model

Andrey Veprikov, Anton Khritankov

66th All-Russian Scientific Conference of MIPT, 2024

A Mathematical Model of the Feedback Loop Effect in Artificial Intelligence Systems

Andrey Veprikov, Anton Khritankov, Alexander Afanasiev

Mathematical Methods for Pattern Recognition (MMPR), 2023

Change-Point Detection via Representation Learning with Labeled and Unlabeled Data

Andrey Veprikov, Alexander Stepikin, Evgenia Romanenkova, Alexey Zaytsev

65th All-Russian Scientific Conference of MIPT, 2023

SELECTED PREPRINTS & PAPERS UNDER REVIEW

Beyond SGD, Without SVD: Proximal Subspace Iteration LoRA with Diagonal Fractional K-FAC

Abdulla Jasem Almansoori, Maria Ivanova, Andrey Veprikov, Aleksandr Beznosikov, Samuel Horvath, Martin Takac

2026

LionMuon: Alternating Spectral and Sign Descent for Efficient Training

Arman Bolatov, Artem Riabinin, Nikita Kornilov, Andrey Veprikov, Samuel Horvath, Martin Takac, Aleksandr Beznosikov

2026

Zero-Order Optimization for LLM Fine-Tuning via Learnable Direction Sampling

Valery Parfenov, Grigoriy Evseev, Andrey Veprikov, Nikolay Bushkov, Stanislav Moiseev, Aleksandr Beznosikov

2026

Where Does Warm-Up Come From? Adaptive Scheduling for Norm-Constrained Optimizers

Artem Riabinin, Andrey Veprikov, Arman Bolatov, Martin Takac, Aleksandr Beznosikov

2026

Faster Than SVD, Smarter Than SGD: The OPLoRA Alternating Update

Abdulla Jasem Almansoori, Maria Ivanova, Andrey Veprikov, Aleksandr Beznosikov, Samuel Horvath, Martin Takac

2025

Aligning Distributionally Robust Optimization with Practical Deep Learning Needs

Dmitrii Feoktistov, Igor Ignashin, Andrey Veprikov, Nikita Borovko, Alexander Bogdanov, Savellii Chezhegov, Aleksandr Beznosikov

2025

Preconditioned Norms: A Unified Framework for Steepest Descent, Quasi-Newton and Adaptive Methods

Andrey Veprikov, Arman Bolatov, Aleksandr Bogdanov, Samuel Horvath, Aleksandr Beznosikov, Martin Takac, Slavomir Hanzely

2025